



Research review paper

Amino acid misincorporation in recombinant proteins

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ABSTRACT

Proteins provide the molecular basis for cellular structure, catalytic activity, signal transduction, and molecular transport in biological systems. Recombinant protein expression is widely used to prepare and manufacture novel proteins that serve as the foundation of many biopharmaceutical products. However, protein translation bioprocesses are inherently prone to low-level errors. These sequence variants caused by amino acid misincorporation have been observed in both native and recombinant proteins. Protein sequence variants impact product quality, and their presence can be exacerbated through cellular stress, overexpression, and nutrient starvation. Therefore, the cell line selection process, which is used in the biopharmaceutical industry, is not only directed towards maximizing productivity, but also focuses on selecting clones which yield low sequence variant levels, thereby proactively avoiding potentially inauspicious patient safety and efficacy outcomes. Here, we summarize a number of hallmark studies aimed at understanding the mechanisms of amino acid misincorporation, as well as exacerbating factors, and mitigation strategies. We also describe key advances in analytical technologies in the identification and quantification of sequence variants, and some practical considerations when using LC-MS/MS for detecting sequence variants.

1. Introduction

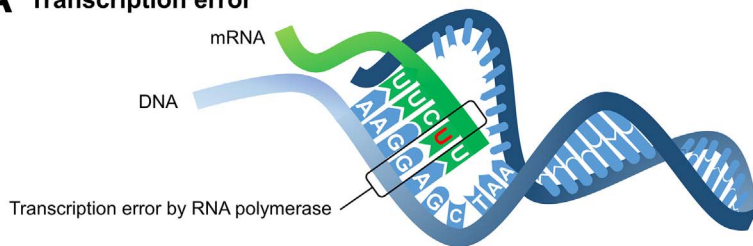
Recombinant protein expression is used to produce protein therapeutics, including monoclonal antibodies, Fc-fusion proteins, enzymes, and peptides. The expression system and culturing conditions can influence protein product quality attributes, such as translational fidelity and post-translational modifications, which can in turn impact therapeutic efficacy and patient safety. In this regard, protein quality attributes are of utmost importance for the biopharmaceuticals industry. Consequently, biopharmaceuticals are governmentally regulated, and safeguards exist to ensure product efficacy and patient safety. While fairly robust, recombinant protein expression is not infallible. Amino acid misincorporation is the errant replacement of an amino acid by another during the translation process, producing unintended protein sequence variants. Amino acid misincorporation inherently occurs, generally at low levels, at an average baseline frequency of 10^{-6} to 10^{-3} , depending on the expression system and cell line (Bouadloun et al., 1983; Edelmann and Gallant, 1977; Ellis and Gallant, 1982; Kramer and Farabaugh, 2007; Kramer et al., 2010; Laughrea et al., 1987; Loftfield, 1963; Loftfield and Vanderjagt, 1972; Manickam et al., 2014; Stansfield et al., 1998; Yu et al., 2009; Zhang et al., 2013). Among the unintended sequence variants caused by amino acid misincorporation, it is estimated that 10% to 50% will affect protein function (Drummond and Wilke, 2008, 2009). Sequence variants can

cause protein heterogeneity, alter catalytic activity (Kramer and Farabaugh, 2007; Stansfield et al., 1998), disrupt ligand/substrate binding, and perturb protein folding leading to protein aggregation (Drummond and Wilke, 2008; Lee et al., 2006). Unintended substitution of the amino acids (Asn, Ser, Thr) in N-linked and O-linked glycosylation recognition sequences can also impact in-serum half-life, protein structure, therapeutic efficacy, and antibody-dependent cellular cytotoxicity (Butler and Spearman, 2014; Wright et al., 1990). Therefore, an important consideration for biopharmaceuticals is that some high-level sequence variants can potentially induce undesired immune response in patients through sequence heterogeneity and increased aggregation (Drummond and Wilke, 2009). In many cases, sequence variants will be relatively innocuous. Nonetheless, the impact of a specific sequence variant on product quality is often difficult to predict. Even more challenging to assess, in a pre-clinical setting, is the way a product attribute change accompanying sequence variants may affect individual patients (Singh, 2011). Therefore, production cell line selection, a key component of biopharmaceuticals development, includes considerations and workflows to minimize sequence variants as a precaution against potentially deleterious clinical and product outcomes. Sequence variant analysis of protein products is routinely performed across the biopharmaceutical industry. Many observed sequence variants are also caused by DNA mutations. While important, DNA mutations will not be extensively discussed in this review.

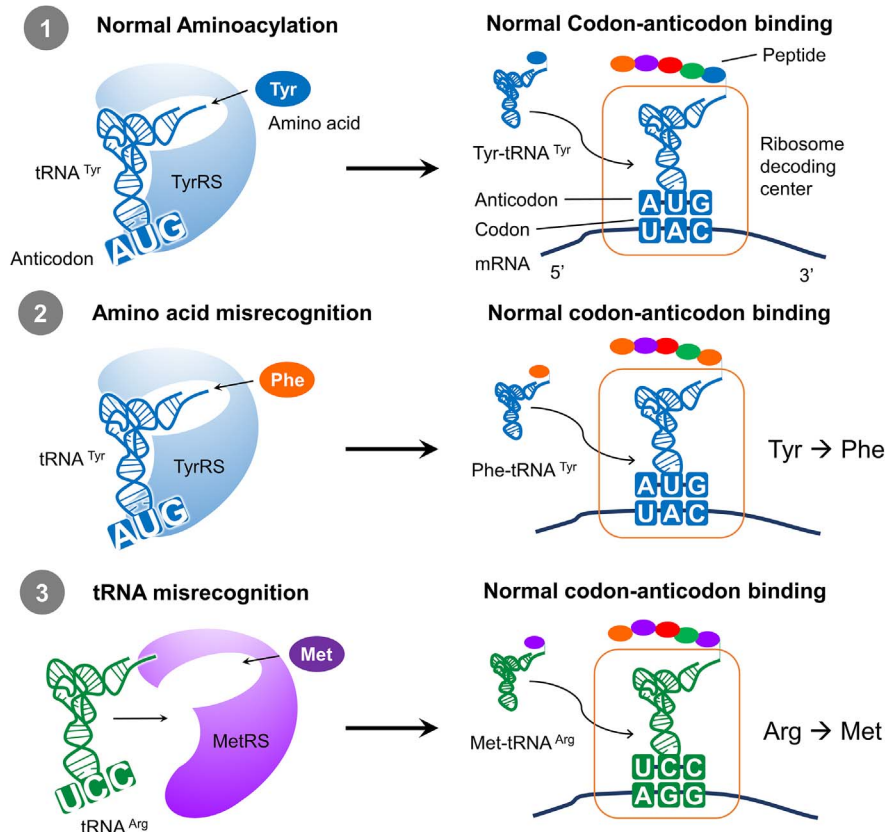
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A Transcription error



B Misacylation



C Codon-anticodon mispairing

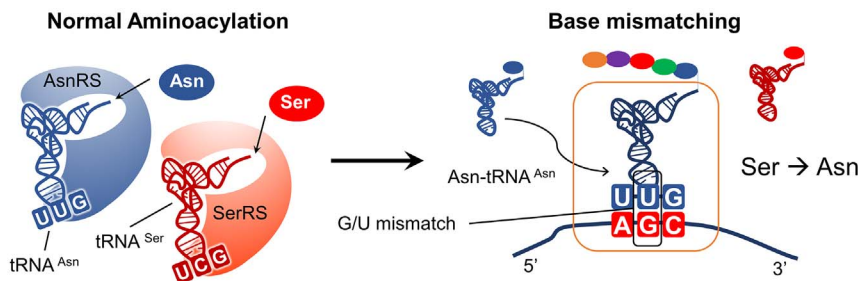


Fig. 1. Amino acid misincorporation mechanisms. Panel A depicts a transcriptional error caused by RNA polymerase. Here G pairs with a U instead of a C leading to a C → U base change. Panel B depicts normal translation and the major misacylation mechanisms. 1) Aminoacyl-tRNA synthetase (aaRS) causes aminoacylation of cognate tRNA (left), followed by codon-anticodon pairing (right), leading to the elongation of the peptide with the correct amino acid sequence. 2) AaRS (TyrRS) misrecognizes a noncognate amino acid (Phe) forming Phe-tRNA^{Tyr} (left), followed by a normal codon-anticodon pairing (right), leading to a Tyr → Phe misincorporation. 3) AaRS (MetRS) misrecognizes a noncognate tRNA (tRNA^{Arg}) forming Met-tRNA^{Arg} (left), followed by normal codon-anticodon pairing (right), leading to an Arg → Met misincorporation. Panel C depicts normal aminoacylation of two tRNAs (tRNA^{Asn} and tRNA^{Ser}) by their corresponding aaRSs (left). The tRNA^{Asn} incorrectly deciphers the Ser codon, through a G/U mismatch in the codon-anticodon pair (right), resulting in a Ser → Asn misincorporation.

Although amino acid misincorporation was first identified in the 1960's, recent advancements in analytical technologies have enabled large-scale, high-resolution, and sensitive sequence variant characterization (Borisov et al., 2015; Brady et al., 2015; Harris and Kilby, 2014; Harris et al., 2012; Hutterer et al., 2012; Ragionieri et al., 2015; Traylor et al., 2016; Yang et al., 2010; Yu et al., 2009; Zeck et al., 2012; Zhang et al., 2013). This has deepened insight of amino acid misincorporation mechanisms, their exacerbating factors, their impact on product

quality, and means to counteract the mechanisms. Herein, we describe a number of hallmark studies geared towards understanding amino acid misincorporation for various protein products across each of the major protein expression systems. Additionally, we discuss key analytical technologies that have afforded sensitive detection and characterization of amino acid misincorporation in protein products.

2. Overview of protein translation

Proteins provide the molecular basis for cellular structure, catalytic activity, signal transduction, and transport that are necessary for biological function. Protein expression is the process by which proteins are biosynthesized from amino acid building blocks. Key steps in protein translation are the same in all recombinant protein expression systems, including mammalian, microbial, yeast and insect cells. When protein expression is initiated, gene-encoding DNA is first transcribed by RNA polymerase into mRNA, the protein translation blueprint. After transcription, the ribosome-binding site of the mRNA solicits ribosome assembly around the start codon. Simultaneously, amino acids are sequestered and covalently charged to their cognate tRNA by their respective aminoacyl-tRNA synthetase (aaRS) to form aminoacyl-tRNAs (aa-tRNA) at the expenditure of ATP. The aa-tRNAs, which are now ready for translation, are transported to the ribosome by elongation factors. For both chain initiation and elongation, the protein sequence encoded in the mRNA is deciphered by the tRNA through complementary codon-anticodon pairing. The new amino acid is added to the carboxyl-terminus of the elongating peptide chain at the expense of GTP. Peptide chain elongation continues until a termination signal, such as a stop codon, is reached, which initiates release factors that disassemble the translational machinery. Both native and recombinant protein expression employ the same set of cellular machinery (Butler and Meneses-Acosta, 2012; Lai et al., 2013; Ling et al., 2009; Schmeing and Ramakrishnan, 2009; Zhu, 2012).

3. Mechanisms of amino acid misincorporation

While the fidelity of protein biosynthesis is crucial for proper cellular function, all biological systems, including protein translation, can produce errors. Amino acid misincorporation inherently occurs at low levels, which leads to unintended sequence variants. Since recombinant protein expression relies on the endogenous host biosynthetic machinery (tRNAs, aaRSs, ribosomes, and elongation factors), it is susceptible to the same translation error mechanisms (Harris and Kilby, 2014). Amino acid misincorporation in native human serum albumin, resulting under normal physiological conditions, has been measured on the order of 10^{-5} and represents a true baseline mammalian cell misincorporation level (Zhang et al., 2013). The baseline frequency of sequence variants in recombinant proteins is on the order of 10^{-5} to 10^{-3} , and is representative of misincorporation in a production environment where some exacerbating factors, such as fast cell growth and high protein productivity are likely to be present (Harris and Kilby, 2014; Zhang et al., 2013). Furthermore, error rates can dramatically increase under cellular stress, overexpression, fast growth, nutrient starvation, and in the presence of mutagens and reactive oxygen species. Amino acid misincorporation can arise from faults during the three information propagation points in protein synthesis, including 1) transcription, 2) tRNA charging (aminoacylation), and 3) codon-anticodon pairing (Gingold and Pilpel, 2011; Reynolds et al., 2010; Rozov et al., 2016a; Zaher and Green, 2009). Fig. 1 depicts each of the major amino acid misincorporation mechanisms.

3.1. Transcription error

Found in all organisms, RNA polymerase is responsible for transcription of gene-encoding DNA into mRNA. Despite having some proofreading and editing functions, RNA polymerase inherently produces errors at a rate of 10^{-6} to 10^{-4} per base (Gout et al., 2013; Traverse and Ochman, 2016). Faulty base recognition and defective transcription editing domains have been shown to cause high transcriptional error rates leading to the loss of proteostasis and a shortened cellular lifespan (Vermulst et al., 2015). Traverse and Ochman determined that *E. coli* transcription error rates are mostly on the order of 5×10^{-6} , with the exception of the C $\rightarrow$ U base change, which has a

six-fold higher error frequency of approximately 3×10^{-5} (Traverse and Ochman, 2016). Interestingly, in earlier work, Zhang et al. (2013) analyzed native human serum albumin purified from three different subjects, and observed numerous sequence variants with error rates on the order of 1×10^{-5} to 1×10^{-4} (a total of 66 observed, involving 15 types of amino acid changes across 28 different codons). Among the observed sequence variants, one-third corresponded to a C $\rightarrow$ U base change, and exhibited error frequencies from 2×10^{-5} to 1×10^{-4} . Zhang et al. attributed these sequence errors to G/U mismatching during transcription, which is consistent with and validated by the C $\rightarrow$ U transcriptional error rates determined by Traverse and Ochman. Sequence variants corresponding to C $\rightarrow$ U base changes were also observed in recombinant proteins expressed in *E. coli* with similar error rates (Zhang et al., 2013). Transcriptional errors are not extensively reported due to their relatively low error frequencies. When over-expressing proteins, sequence variants caused by transcriptional error (up to 1×10^{-4}) (Traverse and Ochman, 2016) are usually less prevalent when compared to those resulting from mistranslation occurring after transcription (up to 1×10^{-3}) (Harris and Kilby, 2014; Zhang et al., 2013). For native human proteins, however, the error rates of the two processes are similar (both up to 1×10^{-4}) (Zhang et al., 2013). Transcription error is depicted in Fig. 1A.

3.2. tRNA mischarging

Aminoacyl-tRNA synthetases (aaRS) are a set of enzymes that bridge the interface between nucleic acids and proteins, playing an indispensable role in functionalizing the amino acid and tRNA pools for protein translation. AARSs rely on substrate specificity and editing capability to direct high-fidelity tRNA aminoacylation (Reynolds et al., 2010). Known as tRNA misacylation or mischarging, incorrect amino acids can, however, be charged to non-cognate tRNA due to imperfect aaRS specificity and/or editing function (Ling et al., 2009). Once formed, many mischarged aminoacyl-tRNA can then be efficiently and indiscriminately used by the ribosome (Effraim et al., 2009). As such, ATP expenditure during aminoacylation is considered a major translation error checkpoint (Reynolds et al., 2010; Yadavalli and Ibba, 2013). Misacylation has been reported on the order of 10^{-5} to 10^{-3} (Feeney et al., 2013). And in in vitro studies, misacylation, measured by ATP-pyrophosphatase transfer activity, occurs at similar levels for major misacylation products (Jakubowski, 1999; Jakubowski and Fersht, 1981; Jakubowski and Goldman, 1992). The major misacylation mechanisms are shown in Fig. 1B.

Interestingly, aaRS substrate specificity and editing capabilities are not evolutionarily conserved, and can vary between species (Yadavalli and Ibba, 2013). For example, mammalian tyrosyl-tRNA synthetase (TyrRS) was found to be more error-prone compared to the bacterial counterpart (Raina et al., 2014). A number of studies also highlight the importance of aaRS fidelity and editing function. Mutations compromising the editing domain of alanyl-tRNA synthetase (AlaRS) have been shown to increase misacylation by Gly and Ser, leading to host cell disease pathology (Lee et al., 2006). For biotechnology applications, naturally occurring low-fidelity phenylalanyl-tRNA synthetases (PheRS) and mutant PheRS have both been exploited to incorporate non-natural amino acids during protein translation (Noren et al., 1989). Additionally, cellular stress can promote misacylation. PheRS can mischarge naturally occurring cellular oxidation byproducts such as *m*-Tyr (Bullwinkle et al., 2014; Klipcan et al., 2009). Mitochondrial respiration is a major source of the reactive oxygen species responsible for many oxidation reactions and for a large pool of non-proteinogenic amino acids (outside of the standard 20 genetically encoded amino acids) (Stadtman and Levine, 2003). Subsequent studies have shown that the misincorporation of many non-proteinogenic amino acids occur by misacylation of the modified amino acids, rather than through a post-translational oxidation of the native amino acid in the peptide sequence (Bullwinkle and Ibba, 2016). The oxidation of the cellular

amino acid pool can lead to cytotoxic mistranslation (Bullwinkle et al., 2014). In vitro misacylation studies have revealed that structurally similar proteinogenic and non-proteinogenic amino acids are often more likely than dissimilar amino acids to outcompete the cognate amino acid for charging. For example, norvaline and homocysteine were observed to readily replace leucine and methionine, respectively, in in vitro studies (Chen et al., 2011; Cveticic et al., 2014; Jakubowski and Fersht, 1981).

Alternatively, in some cases, for example in AspRS-deficient organisms, AsnRS functions to aminoacylate both tRNA^{Asp} and tRNA^{Asn}, which demonstrates that flexibility in aaRS specificity can be essential for cellular function and also supports the idea that aaRSs often have context-dependent specificity and editing (Bullwinkle et al., 2014). Work by Momand and Clarke also showed that in *E. coli* aspartic acid can attach at low-levels to tRNA^{Asp} through the side chain β -carboxyl group rather than the normal α -carboxyl linkage, which leads to the misincorporation of isoaspartyl and a kinked peptide backbone structure (Momand and Clarke, 1990). Additionally, tRNAs contain recognition elements in the form of specific nucleotide sequences and modifications that mediate recognition by the aaRS. Invariably when this recognition fails, errors occur. For example, even when MetRS recognizes its cognate amino acid, methionine, it can proceed to mis-methionylate tRNA^{Arg} and tRNA^{Lys} leading to the Arg $\rightarrow$ Met and Lys $\rightarrow$ Met misincorporations (Jones et al., 2011; Ling and Soll, 2010; Netzer et al., 2009; Pan, 2013; Wiltout et al., 2012). It is believed that under oxidative stress, the cells increase incorporation of methionine to combat reactive oxygen species (Netzer et al., 2009). While seemingly detrimental, growing evidence also suggests that these examples of aaRS infidelity may be a part of a set of adaptive translation mechanisms to improve survival under stress (Pan, 2013; Ribas de Pouplana et al., 2014).

3.3. Codon-anticodon mispairing

Codon-anticodon pairing (Fig. 1C) functions to decode the protein sequence information contained in mRNA. Missense amino acid replacements can result from non-complementary codon-anticodon pairing inside the ribosome. The most common deviations in codon-anticodon pairing that cause sequence variants are G/U and third-position wobble mismatches (Manickam et al., 2014; Rozov et al., 2016a; Zhang et al., 2013). This propensity explains many of the commonly observed sequence variants in recombinant proteins expressed in both microbial and mammalian cells, and was further used to successfully predict sequence variant products for various codons (Zhang et al., 2013). Since base mispairing is thermodynamically unfavorable, the ribosome can sterically discriminate against many errors, especially at the first codon position (Rozov et al., 2016b). Nonetheless, the G/U base pair can adopt a configuration that is isometric to canonical Watson-Crick pairs, which facilitates entry and usage in the ribosome. The follow-up study revealed that this structural transformation occurs by a keto-enol tautomerization (at an estimated rate of 10^{-4}) (Rozov et al., 2015, 2016b; Westhof, 2014). In some cases, the ribosome can even sterically facilitate conversion of G/U mismatches to Watson-Crick geometry (Rozov et al., 2015, 2016b). Interestingly, G/U pairing in RNA has exhibited biological relevance, playing essential chemical, structural, and ligand-binding roles (Varani and McClain, 2000). While less common, U/U and U/C base mispairings also lead to faulty codon recognition (Manickam et al., 2014; Zhang et al., 2013), presumably by adopting canonical-like geometry.

4. Amino acid misincorporation in different expression systems

Amino acid misincorporation was first reported as early as the 1960's in proteins produced in animal as well as microbial cells (Edelmann and Gallant, 1977; Loftfield, 1963; Loftfield and Vanderjagt, 1972; Palmer and Wilhelm, 1978; Parker et al., 1978). Aided by

advancements in analytical technologies, most notably mass spectrometry, sequence variants have been confirmed and quantified, both in native and recombinant proteins, produced in *E. coli*, yeast, and mammalian cells (Harris and Kilby, 2014; Harris et al., 2012; Hutterer et al., 2012; Ragionieri et al., 2015; Traylor et al., 2016; Zeck et al., 2012; Zhang et al., 2013). While insect cells are actively used for recombinant protein expression, incidences of amino acid misincorporation have yet to be reported, likely owing to slower mainstream adoption of the technology (van Oers et al., 2015).

4.1. Microbial expression systems

For microbial recombinant protein expression, amino acid misincorporation frequency per residue can range from 0.001% and have been reported up to 42% depending on the codon being used and fermentation conditions (Apostol et al., 1997; Calderone et al., 1996; Forman et al., 1998; Harris et al., 2012; Huang et al., 2012; Hutterer et al., 2012; Kane, 1995; Kramer and Farabaugh, 2007; Kramer et al., 2010; Lu et al., 1993, 1988, 1995; Manickam et al., 2014; McNulty et al., 2003; Randhawa et al., 1994; Sandbaken and Culbertson, 1988; Schneider et al., 2005; Scorer et al., 1991; Seetharam et al., 1988; Stansfield et al., 1998; Zhang et al., 2013). In *E. coli*, Ala, Asp, Glu, Phe, Gly, His, Leu, Met, Asn, Pro, Gln, Arg, Ser, Val, and Tyr residues are prone to misincorporation (Table 1) (Apostol et al., 1997; Bogosian et al., 1989; Calderone et al., 1996; Forman et al., 1998; Harris et al., 2012; Huang et al., 2012; Lu et al., 1993, 1988; McNulty et al., 2003; Randhawa et al., 1994; Schneider et al., 2005; Seetharam et al., 1988; Tsai et al., 1988; Yu et al., 2009; Zhang et al., 2013). Significantly higher rates of translational infidelity, above baseline levels, however, have often been linked to single and clustered rare codons in the mRNA coding sequence (Calderone et al., 1996; Forman et al., 1998; Kane, 1995; McNulty et al., 2003; Seetharam et al., 1988). Because mammalian cells and microbial cells have differences in codon usage, when mammalian genes are expressed in microbial cells, the presence of rare codons may affect the translational efficiency and fidelity (Kane, 1995). The most comprehensively studied are the set of Arg rare codons (AGA, AGG, and CGG). Arg rare codons can cause varying severities of frameshift, product truncation (from premature translation stoppage), and unintended substitution by Lys and Gln (Calderone et al., 1996; Kane, 1995; McNulty et al., 2003; Seetharam et al., 1988). The truncated and frameshift sequence variants have been attributed to ribosome stalling, a unique prokaryotic response to cellular substrate deficiencies, and ribosome slipping, an inadvertent skip in the mRNA nucleotide read, respectively (Gurvich et al., 2005). Other rare codons such as GGA (Gly), CUA (Leu), and CCC (Pro) have also exhibited similar erroneous outcomes (Huang et al., 2012; Kane, 1995). Codon optimization strategies, preferably employing amino acid misincorporation data (Hutterer et al., 2012) are therefore necessary for efficient and high-quality human therapeutics production from microbial cells.

While codon usage is an important factor that dictates misincorporation propensity, in vitro studies have shown that among all aa-tRNAs, the tRNA codon recognition kinetic rate constants cover a fairly narrow range (100–140 mM aa-tRNA⁻¹ s⁻¹), and are first-order with respect to aa-tRNA concentration (Dittmar et al., 2005; Gromadski et al., 2006; Kramer and Farabaugh, 2007). Additionally, Kramer and colleagues showed that a subset of near-cognate tRNA misread (one base-pair mismatch) occurred at an order of magnitude higher frequency than other near- and non-cognate misreads. The difference in tRNA codon recognition rates suggests that amino acid misincorporation is largely determined by tRNA pairing competition (Kramer and Farabaugh, 2007). As a result, overexpression of Arg rare codon tRNAs, which are normally found at relatively low concentrations, was able to significantly decrease Arg $\rightarrow$ Lys misincorporation rates (Calderone et al., 1996). Similarly, replacing rare (AGG and CGG) with a high-usage Arg codon (CGU) successfully reduced Arg $\rightarrow$ Gln variants from 0.08% to undetectable levels (Hutterer et al., 2012).

Table 1
Amino acid misincorporation products observed in recombinant proteins expressed in microbes.

Misincorporation	Codon	Expression system	Proposed mechanism	Methodology ^a	Reference ^b
A → T	GCC; GCU	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1
D → E	GAC; GAU	<i>E. coli</i>	Wobble mismatch	LC-MS/MS	1, 2
E → K	GAA	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1, 2
F → I/L	n/a	<i>E. coli</i>	n/a	LC-MS/MS	2
G → D	GGC; GGU	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1, 3
G → E	GGA; GGG	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1, 3, 4
G → R	GGA	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1
G → S	GGU; GGC	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1
H → Q	CAC; CAU	<i>E. coli</i>	Wobble mismatch	Edman; LC-MS/MS	1, 5, 6
K → R	n/a	<i>E. coli</i>	n/a	LC-MS/MS	3
L → K	n/a	<i>E. coli</i>	n/a	LC-MS/MS	2
L → norvaline	CUA; CUG	<i>E. coli</i>	tRNA mischarging	Edman	7
L → V	n/a	<i>E. coli</i>	n/a	LC-MS/MS	2
M → I/L	AUG	<i>E. coli</i>	Wobble mismatch	LC-MS/MS	1
M → norleucine	AUG	<i>E. coli</i>	n/a	Edman; AAA	8, 9, 10, 11
M → T	AUG	<i>E. coli</i>	U/G mismatch	LC-MS/MS	1
N → K	AAC; AAU	<i>E. coli</i>	Wobble mismatch	LC-MS/MS	1
P → L/I	CCC	<i>E. coli</i>	Transcriptional G/U mismatch	LC-MS/MS	1
P → S	CCA; CCC; CCG	<i>E. coli</i>	Transcriptional G/U mismatch	LC-MS/MS	1
Q → H	CAA	<i>E. coli</i>	Wobble mismatch	LC-MS/MS	1
R → K	AGA; AGG	<i>E. coli</i>	G/U mismatch	LC-MS/MS; Edman	1, 12, 13, 14
R → Q	CGA; CGG	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1, 3, 15
S → N	AGC; AGU	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1, 16
S → T	n/a	<i>E. coli</i>	n/a	LC-MS/MS	2
V → I/L	GUC; GUU	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1
V → M	GUG	<i>E. coli</i>	G/U mismatch	LC-MS/MS	1
Y → N	UAC	<i>E. coli</i>	U/U mismatch	LC-MS/MS	1, 3
Stop → W	UGA	<i>E. coli</i> , yeast	Wobble mismatch	Edman; LC-MS/MS	17, 18

^a Edman: Edman sequencing; AAA: amino acid analysis.

^b 1. (Zhang et al., 2013); 2. (Harris et al., 2012); 3. (Hutterer et al., 2012); 4. (Huang et al., 2012); 5. (Lu et al., 1993); 6. (Schneider et al., 2005); 7. (Apostol et al., 1997); 8. (Lu et al., 1988); 9. (Tsai et al., 1988); 10. (Bogossian et al., 1989); 11. (Randhawa et al., 1994); 12. (Seetharam et al., 1988); 13. (Calderone et al., 1996); 14. (Forman et al., 1998); 15. (McNulty et al., 2003); 16. (Yu et al., 2009); 17. (Lu et al., 1995); 18. (Meng et al., 1995).

Stop codon readthrough, in particular, highlights and substantiates the significance of tRNA binding competition as a key determinant of misincorporation. In recombinant methionyl-neurotrophin and glutathione S-transferase expressed in microbes, Gln, Tyr and Lys have been shown to insert at UAA (ochre) and UAG (amber) stop codons, and Trp, Cys, and Arg can insert at the UGA (opal) stop codon (Blanchet et al., 2014; Lin et al., 1986; Meng et al., 1995). Given the efficiency of the opal stop codon, opal readthrough was attributed to competition between tRNA^{Trp} and release (termination) factors (Lu et al., 1995; Meng et al., 1995). It was later revealed that two different glutamine tRNA^{Gln} were responsible for Gln insertion at UAA and UAG stop codons in *S. cerevisiae* (Blanchet et al., 2014; Lin et al., 1986).

4.2. Mammalian expression systems

Chinese hamster ovary (CHO) cells are extensively used for recombinant protein expression and account for the production of over 65% of the biopharmaceuticals manufactured (Kim et al., 2012). Significant investments have been made towards improving the translational productivity and fidelity of mammalian expression systems. There is also invigorated interest in understanding the mechanisms of amino acid misincorporation in mammalian cells (Feeney et al., 2013; Guo et al., 2010; Khetan et al., 2010; Raina et al., 2014; Yu et al., 2009; Zhang et al., 2013). Many amino acid misincorporation products in recombinant proteins expressed in mammalian cells have been reported (Table 2). As cell culture substrate and nutrient concentrations can impact misincorporation levels, a number of studies have investigated the importance of these parameters. Khetan and colleagues identified Asn → Ser misincorporation in recombinant IgG mAb at a frequency of 2×10^{-4} to 2×10^{-3} , where substitution levels corresponded with the depletion of extracellular Asn below 0.1 mM (Khetan et al., 2010; Wen et al., 2009). Although misincorporation was only observed at low Asn concentrations, substitution levels could be exacerbated further by

altering the Ser:Asn ratio by increasing Ser concentration (Khetan et al., 2010). Yu and colleagues also identified Ser → Asn misincorporation in mAbs expressed in CHO cells (Yu et al., 2009). Although Ser is encoded by six redundant codons, misincorporation of Asn at Ser sites was only detected at the AGC codon in that work. Replacing the AGC codon with UCC and UCU was found to eliminate the Ser → Asn misincorporation (Guo et al., 2010). Furthermore, the Ser → Asn misincorporation rate varied depending on the Ser location and on the surrounding coding sequence (Yu et al., 2009), which suggests that mRNA secondary structure and kinetic control may play a role in substitution propensity (Shpaer, 1985). A nutrient deprivation study conducted by Feeney and colleagues as well as Raina et al. also revealed the promiscuous replacement of Tyr by Phe and, to a lesser degree, by His (Feeney et al., 2013; Raina et al., 2014). Tyr → Phe and Tyr → His variants were detectable when extracellular Tyr concentrations decreased below 0.02 mM. By varying starvation conditions, tRNA^{Tyr} misacylation by Phe was established to be the likely culprit (Feeney et al., 2013; Raina et al., 2014).

Recently Zhang and colleagues completed an extensive characterization of amino acid misincorporation in various recombinant therapeutics expressed in CHO cells and *E. coli*. Quantifiable amino acid misincorporation levels were detected by high-resolution tandem mass spectroscopy at Gly, Met, Asn, Asp, Ser, and Val residues in CHO-expressed IgG mAbs. Ultimately, Zhang and colleagues uncovered a strong link (45% of study cases) between the instances of amino acid misincorporation and G/U codon-anticodon base-pair mismatches in both CHO and *E. coli* expression systems (Zhang et al., 2013). These observations were then used to reliably predict cases with a high-level misincorporation propensity in *E. coli* (Zhang et al., 2013). These results are consistent with structural elucidation of codon-anticodon pairing within the 70S ribosome showing that the G/U mismatch can adopt Watson-Crick pair isometry through a keto-enol tautomeric shift to facilitate misincorporation (Rozov et al., 2015, 2016; Varani and

Table 2

Amino acid misincorporation products observed in recombinant proteins expressed in mammalian cells.

Misincorporation	Codon	Expression system	Proposed mechanism	Methodology	Reference ^a
A → T	n/a	Murine	n/a	LC-MS/MS	1
D → G	GAC	CHO	Transcriptional G/U mismatch	LC-MS/MS	2
D → N	GAC	CHO	G/U mismatch	LC-MS/MS	2
F → m-, o-tyrosine	n/a	CHO	tRNA mischarging	LC-MS/MS	3
G → D	GGC; GGU	CHO, murine	G/U mismatch	LC-MS/MS	1, 2
K → R	n/a	Murine	n/a	LC-MS/MS	1
M → I/L	AUG	CHO	Wobble mismatch	LC-MS/MS	2
N → K	AAC; AAU	CHO, murine	Wobble mismatch	LC-MS/MS	1, 2, 4
N → S	AAC; AAU	CHO	tRNA mischarging	LC-MS/MS	2, 5
R → G	n/a	Murine	n/a	LC-MS/MS	1
R → K	n/a	Murine	n/a	LC-MS/MS	1
S → N	AGC; AGU	CHO; murine	G/U mismatch	LC-MS/MS	1, 2, 6
S → R	AGC; AGU	CHO	Wobble mismatch	LC-MS/MS	1, 2
T → S	n/a	Murine	n/a	LC-MS/MS	1
V → I/L	GUC	CHO, murine	G/U mismatch	LC-MS/MS	1, 2
Y → F	UAU; UAC	CHO	tRNA mischarging	LC-MS/MS	7, 8
Y → H	n/a	CHO	tRNA mischarging	LC-MS/MS	8

^a 1. (Borisov et al., 2015); 2. (Zhang et al., 2013); 3. (Popp et al., 2015); 4. (Wen et al., 2009); 5. (Yu et al., 2009); 6. (Ren et al., 2011); 7. (Raina et al., 2014); 8. (Feeney et al., 2013).

McClain, 2000).

4.3. Native proteins

Amino acid misincorporation in native proteins has not been extensively studied (Table 3). To our best knowledge, the investigation by Zhang and colleagues of native human serum albumin (Zhang et al., 2013), was the first, to reveal a pattern of sequence variants in human native protein. Sequence variants in native human proteins were found to be lower than overexpressed recombinant proteins, approaching transcriptional error rates (Traverse and Ochman, 2016). In fact, one-third of observed misincorporation in human serum albumin was attributed to transcriptional error (Zhang et al., 2013). These observations from native human proteins may serve as a guide for establishing safe sequence variant levels in therapeutic proteins (Harris and Kilby, 2014).

5. Strategies for eliminating sequence variants

In order to minimize protein sequence variants, strategies have been proposed and implemented to counteract specific misincorporation mechanisms at the genetic, translation, and cellular levels.

5.1. Expression construct design and codon optimization

Significant opportunities for minimizing amino acid misincorporation are available at the DNA level. After elucidating the link between rare codons and unintended amino acid substitution, targeted replacement of Arg rare codons (CGG, AGG) with a higher-usage codon (CGU), in combination with appropriate culture conditions, successfully reduced the frequency of Arg → Gln misincorporation in an Fc-fusion protein produced in *E. coli* (Hutterer et al., 2012). Since codon optimization has also been used to promote ribosome-RNA stability and to improve translation productivity, these results highlight an opportunity to simultaneously address translational efficiency and fidelity (Gingold and Pilpel, 2011; Sorensen et al., 1989). A targeted codon optimization strategy, similar to the one employed with *E. coli*, could in principle also be applied to prevent amino acid misincorporation in CHO (Guo et al., 2010; Yu et al., 2009; Zhang et al., 2013). Co-expression or over-expression of key translational elements is also a viable strategy for reducing amino acid misincorporation. In *E. coli*, where rare Arg codons have been implicated in misincorporation, translation infidelity was mitigatable through polycistronic co-expression of the cognate tRNAs, ArgU (tRNA^{Arg}_{UCC}) and/or ArgX (tRNA^{Arg}_{UGG}) (Calderone et al., 1996; McNulty et al., 2003). Expression construct design and codon optimization strategies for mitigating misincorporation are shown in Fig. 2.

Table 3

Amino acid misincorporation products observed in native protein.

Misincorporation	Codon	Organism	Proposed mechanism	Methodology	Reference ^a
A → T	GCA; GCG	Human	G/U mismatch	LC-MS/MS	1
A → V	GCA; GCC; GCU	Human	Transcriptional G/U mismatch	LC-MS/MS	1
D → E	GAU	Human	Wobble mismatch	LC-MS/MS	1
D → N	GAU	Human	G/U mismatch	LC-MS/MS	1
E → K	GAA; GAG	Human	G/U mismatch	LC-MS/MS	1
F → L/I	UUC; UUU	Human	Wobble mismatch	LC-MS/MS	1
H → Y	CAU	Human	Transcriptional G/U mismatch	LC-MS/MS	1
L → F	CUU	Human	Transcriptional G/U mismatch	LC-MS/MS	1
M → I/L	AUG	Human	Wobble mismatch	LC-MS/MS	1
M → T	AUG	Human	U/G mismatch	LC-MS/MS	1
R → K	AGA; AGG	Human	G/U mismatch	LC-MS/MS	1
S → L/I	UCA; UCG	Human	Transcriptional G/U mismatch	LC-MS/MS	1
T → I/L	ACA; ACC; ACU	Human	Transcriptional G/U mismatch	LC-MS/MS	1
V → A	GUA; GUC; GUU	Human	U/G mismatch	LC-MS/MS	1
V → I/L	GUA; GUC; GUU	Human	G/U mismatch	LC-MS/MS	1

^a 1. (Zhang et al., 2013).

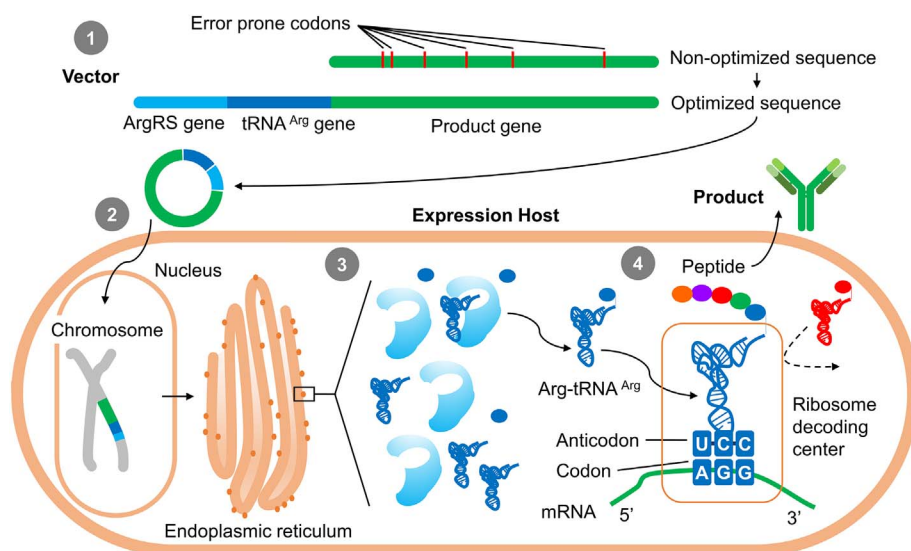


Fig. 2. Strategies for eliminating sequence variants at the genetic and cellular level. 1) The product gene sequence is optimized by removing error-prone codons. The tRNA and aminoacyl-tRNA synthetases for rare codons can be co-expressed. This improves the translation efficiency of rare codons, and in prokaryotes has been shown to mitigate misincorporation. 2) Upon chromosomal integration, the product gene can undergo mutations and cause clonal variability. Therefore, the clone selection process ensures that the clone, which is free of mutated genes, is used for production. The product gene is then transcribed. 3) The mRNA is translated and the tRNA and aminoacyl-tRNA synthetases for the rare codons are co-expressed along with the product. 4) Co-expression can help ensure that the appropriate balance of cognate tRNA and aminoacyl-tRNA synthetases are available to minimize competition from non-cognate charged tRNA. In this example, the tRNA^{Arg} and ArgRS for the rare Arg AGG codon are co-expressed with the product.

5.2. Strain and cell line selection

In the past, biopharmaceutical production CHO cell lines were primarily selected based on viability and productivity. Side-by-side comparison of clones, cultivated under identical process conditions, has shown that sequence variant frequency can depend on the individual clones themselves (Zhang et al., 2013). Clone variability can arise during gene amplification through stochastic transgene integration into the host genome, which can yield an array of genotypes and phenotypes (Kang et al., 2014; Vishwanathan et al., 2015). Stress response, specific productivity, and metabolism, which can directly influence amino acid misincorporation frequency, can consequently be affected (Chong et al., 2012; Kang et al., 2014). Clonal variability can be further worsened by gene silencing in mammalian cells and DNA mutation in both host cell genes and the product genes during transfection (Guo et al., 2010; Harris et al., 1993; Lebkowski et al., 1984; Lee et al., 2006; Pikaart et al., 1998; Sandbaken and Culbertson, 1988; Scott et al., 2014; Traylor et al., 2016; Wan et al., 1999; Zhang et al., 2012, 2013). Mutations to the product gene are an especially important consideration, since product expressed by the cell line will invariably contain unintended sequence variants, with often significantly higher levels that cannot be mitigated in the cell culture process. DNA mutations can be readily distinguished because they are clone-dependent, and appear only at the mutation site, whereas typical amino acid misincorporation mechanisms have a broad effect across the entire protein. A subsequent product DNA and RNA sequencing step can be used to reliably confirm cases where product gene mutation is suspected. Moreover, along with selection for clonal productivity and viability, implementing selection criteria for transgene and product fidelity, and engineering misincorporation-resistant clones can be used as a collective strategy to maximize protein product quality and homogeneity (Harris et al., 1993; Traylor et al., 2016; Zhang et al., 2013, 2015). Strain and cell selection strategies for mitigating misincorporation are shown as part of Fig. 3.

5.3. Process improvements

Another set of successfully demonstrated approaches for tuning translational fidelity and product homogeneity exists on the process development front. As some aminoacyl-tRNA synthetases, such as MetRS and PheRS, are sensitive to reactive oxygen species, an additive such as spermine, which has been proposed to counteract reactive oxygen species (Khan et al., 1992) and has the capability to modulate translation fidelity through tRNA interactions (Naranda and Kucan, 1989), can potentially mitigate MetRS and PheRS misacylation (Loftfield et al.,

1981; Netzer et al., 2009). In time-course studies, it was found that Asn → Ser substitution corresponded with Asn depletion (Khetan et al., 2010; Wen et al., 2009). Therefore, maintaining adequate levels of Asn by controlled feeding was effective at reducing the degree of substitution (Khetan et al., 2010). Also, in cases of Tyr starvation, which resulted in unintended replacement by Phe or His, Tyr supplementation (> 1 mM) prevented the formation of both sequence variants (Feeney et al., 2013). Similar amino acid supplementation strategies have also successfully mitigated amino acid misincorporation in *E. coli* (Dittmar et al., 2005). These findings are promising and reveal that while relatively mature technologies, heterologous *E. coli* and CHO protein expression systems can still benefit from cell culture process improvements. Strategies for mitigating misincorporation during process development are shown in Fig. 3.

Downstream processing may also provide an opportunity to reduce the prevalence of sequence variants in a final product. Products resulting from frameshift, truncation, and stop codon readthrough can often be easily separated by downstream processes due to their significant differences from the desired product. Low-level sequence variants, differing only by one amino acid, can represent a significant challenge for separation technologies, which generally exploit large differences in chemical and physical properties between product and variants. However, many amino acid misincorporation products result in charge changes, such as Asp → Gly, Asp → Asn, Glu → Lys, Gly → Asp, Gly → Glu, Gly → Arg, His → Gln, Leu → Lys, Asn → Lys, Gln → His, Ser → Arg, and Tyr → His. In principle, these variants can be separated from the desired product by charge-based separation techniques such as ion-exchange chromatography (Lu et al., 1993). Having a priori knowledge of sequence variants and their chemical properties may be required to take full advantage of downstream mitigation strategies.

6. Early sequence variant identification methods

Prior to widespread adoption of mass spectrometry (MS), radioisotope labeling combined with electrophoresis was one of the first methods used successfully to identify the presence of sequence variants (Bouadloun et al., 1983; Edelmann and Gallant, 1977; Laughrea et al., 1987; Loftfield, 1963; Loftfield and Vanderjagt, 1972). For recombinant therapeutic proteins, most early observations of sequence variants are based on chromatographic or electrophoretic separation, combined with Edman sequencing and amino acid analysis for identification (Apostol et al., 1997; Bogosian et al., 1989; Lu et al., 1988, 1993, 1995; Meng et al., 1995; Randhawa et al., 1994; Tsai et al., 1988). For

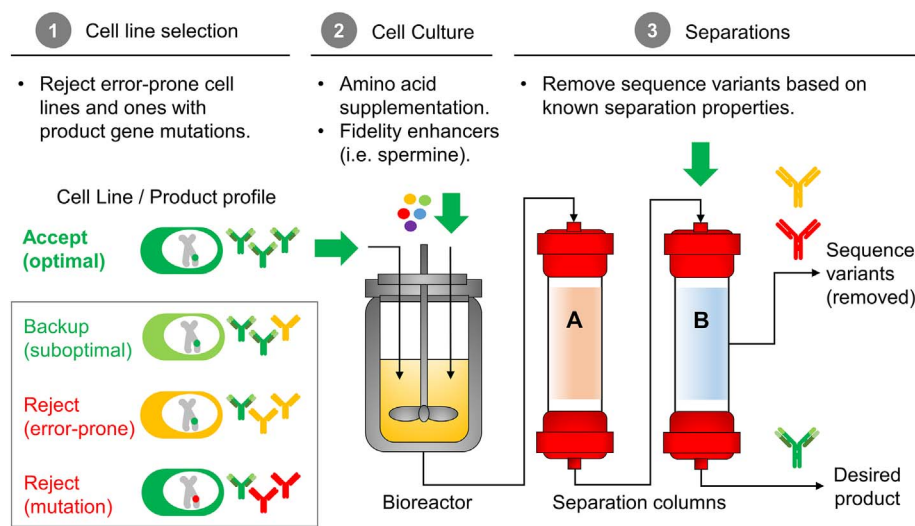


Fig. 3. Strategies for eliminating sequence variants at the process level. 1) Cell line selection is performed to isolate a clone that has minimal sequence variants and generates product that meets product quality attribute requirements. An optimal cell line proceeds further through process development and used in production. A suboptimal cell line may be retained as a backup. Error-prone cell lines and ones with product gene mutations are rejected. 2) Additives such as spermine can be used to improve translation fidelity. Amino acid supplementation can be employed to eliminate cell culture starvation. 3) For downstream process development, additional separation steps can be implemented to remove sequence variants with known properties. For example, ion exchange chromatography can be used when a significant charge change occurs due to an unintended sequence variant (e.g., Asp → Asn, Glu → Lys, His → Gln, etc.).

example, His → Gln substitution was detected in hG-CSF by isoelectric focusing due to a *pI* difference between the wild-type and variant (Lu et al., 1993). While conventional chromatographic and electrophoretic methods are useful for detecting aberrant product attributes involving significant charge changes, frameshift, truncations and extensions, they generally lack the resolution and sensitivity to detect low-level sequence variants.

After the sequence variants of interest were isolated, they were typically identified by Edman sequencing, in which a peptide sequence is obtained by sequential chemical cleavage of the N-terminal residue, using phenyl isothiocyanate, followed by chromatographic identification. The method can accurately sequence up to 50 amino acids at picomole quantities (Laursen, 1971). While the process has been automated, the requirement of a large quantity of purified peptide material is a major limitation. Nonetheless, the method was instrumental for confirming sequence variants caused by stop codon readthrough (Lu et al., 1995; Meng et al., 1995) and by a mutation in the stop codon (Zhang et al., 2012). Furthermore, Edman sequencing can resolve isomers, such as Leu, Ile and norleucine (Bogosian et al., 1989; Lu et al., 1988; Randhawa et al., 1994; Tsai et al., 1988), which are difficult to distinguish with mass spectrometric methods.

Most methods, including mass spectrometry-based methods, detect sequence variants by observing a signal corresponding to the variant. Therefore, reported translational error rates often depend on the detection limit of the methodology. This means that a variant can go unreported if its signal is below the detection limit. Consequently, early reported translational error frequencies are likely overestimated (Manickam et al., 2014) because only the near-cognate, high-abundance misincorporation is reported. An alternative approach is to take advantage of inactive enzyme mutants (Cornut and Willson, 1991; Kramer and Farabaugh, 2007; Kramer et al., 2010; Manickam et al., 2014; Stansfield et al., 1998). For example, Glu537 is required for β -galactosidase enzymatic activity and mutations to this residue abolish enzyme activity. Any observed activity for these mutants, therefore, indicates unintended substitution of residue 537 by glutamic acid. This methodology can then be employed to determine the glutamic acid misincorporation rates for any codon (Manickam et al., 2014). Using this method, mistranslation frequency was determined to be as low as 2×10^{-6} for non-cognate pairing, which is significantly lower than previously believed (Manickam et al., 2014).

7. Identification and quantitation of sequence variants by LC-MS/MS

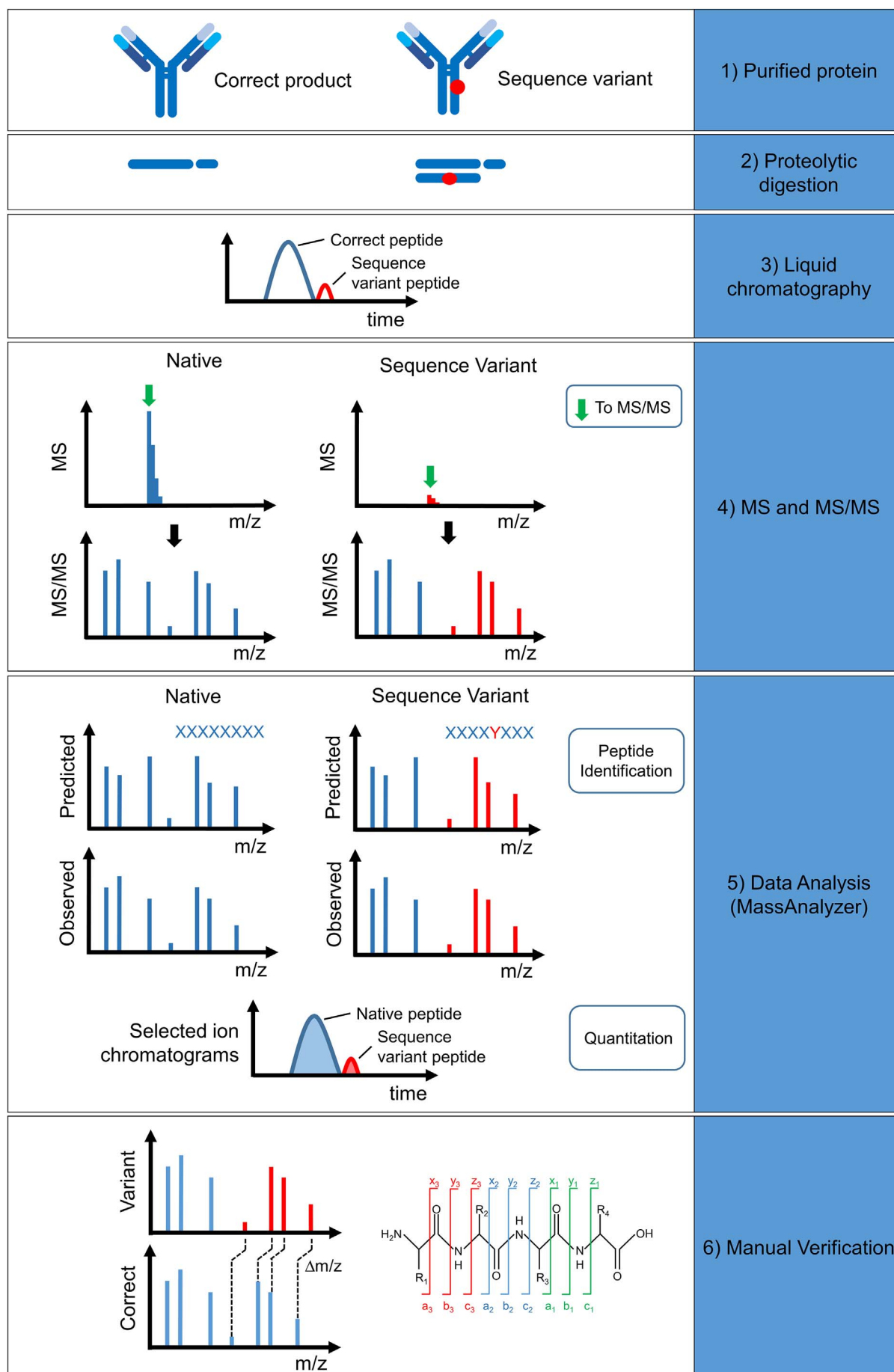
Although historically several different methodologies have been

used to identify protein sequence variants, today most sequence variant analyses are performed on high-resolution LC-MS/MS (liquid chromatography connected with tandem mass spectrometry). Recent advances in mass spectrometry and tandem mass spectrometry, along with informatics tools, have enabled capabilities to routinely monitor large numbers of sequence variants down to the 0.01% range or lower (Borisov et al., 2015; Brady et al., 2015; Harris and Kilby, 2014; Harris et al., 2012; Hutterer et al., 2012; Ragionieri et al., 2015; Traylor et al., 2016; Yu et al., 2009; Zhang et al., 2013).

The current convention for the biopharmaceutical industry is to isolate the product and to characterize sequence variants that are most likely to be retained as part of the final product. Additionally, purifying the product minimizes the bioinformatics and computational burden associated with having to characterize a large population of host cell proteins. To monitor the large number and diversity of sequence variants in a protein, a bottom-up approach, i.e., protein digestion followed by LC-MS/MS analysis, is most commonly employed. Each proteolytic peptide is identified based on its determined mass as well as the sequence information contained in the fragmentation pattern (MS/MS). An identified sequence variant is subsequently quantified based on the intensity of the corresponding peptide ion. Fig. 4 illustrates a typical workflow for LC-MS/MS sequence variant analysis. The following describes this workflow and important considerations for sequence variant analysis.

7.1. Proteolytic digestion

The protein of interest is digested with one or more proteases into short peptides before LC-MS/MS analysis. Prior to digestion, the protein is denatured, reduced, and alkylated to break disulfide bonds, and to ensure that residues are accessible to proteolysis. For proteolysis, trypsin or other high-specificity proteases such as endoproteinase Lys-C, Glu-C, or Asp-N are commonly used. These proteases typically cleave proteins into peptides of appropriate size (500–3000 Da) for MS/MS sequencing. Peptides can inevitably fall outside of this range. Sequence variant identification accuracy decreases dramatically for short peptides (< 5 residues) due to inadequate sequence specificity. Alternatively, long peptides (> 30 residues) often lack sufficient sequence information from the MS/MS to accurately derive the locations of amino acid substitutions. Therefore, proteases can be used in combination to reduce the length of long peptides and improve detection confidence. Separate digestions with different proteases can increase sequence coverage and the number of complementary overlapping peptides for cross-confirmation. While, less specific proteases such as chymotrypsin, thermolysin, pepsin, elastase, etc., are sometimes useful



(caption on next page)

Fig. 4. LC-MS/MS sequence variant analysis workflow. The protein product is first 1) purified, followed by 2) proteolytic digestion. 3) Liquid chromatography separates digested peptides for 4) high-resolution MS and MS/MS analysis. 5) Native and sequence variant peptides are identified by comparing the observed MS/MS to theoretically predicted MS/MS, and the abundance of each sequence variant is determined from peak areas under the selected ion chromatograms. 6) A final manual verification is performed to ensure the accuracy of sequence variant identification.

for sequence variant analysis, they are generally avoided for this application due to poor quantitation accuracy and the frequent occurrence of protease-catalyzed transpeptidation (Schaefer et al., 2005), which may be misinterpreted.

A key consideration for sequence variant analysis is chemical modifications. Modifications can be introduced during sample preparation and some are indistinguishable from amino acid substitutions based on mass, and can be easily misinterpreted. These modifications can also decrease the probability of identifying true sequence variants by drawing mass spectrometer time towards collecting the MS/MS of these chemically modified peptides. Therefore, a fast digestion under relatively mild conditions is important for minimizing these artificial modifications and improving the efficiency of downstream steps (Ren et al., 2009).

The alkylating reagent can also impact sequence variant detection. For example, carboxymethylation using iodoacetic acid produces the same mass change as the Gly → Asp and Ala → Glu substitutions. Because carboxymethylation of methionine, histidine, lysine, and carboxymethylated cysteine occurs to a low extent, when accurate detection of Gly → Asp and Ala → Glu substitutions are important, carbamidomethylation using iodoacetamide is preferred over carboxymethylation.

7.2. Liquid chromatography

Protein digests are first separated by reversed-phase liquid chromatography connected upstream to a mass spectrometer equipped with electrospray ionization interface. For resolving the large number of possible sequence variants, both chromatographic resolution and mass spectral resolution are important. In particular, when the same amino acid substitution occurs at multiple locations on the same peptide, appropriate HPLC resolution can provide the additional information needed to identify and quantify the substitution at each site. Therefore, ultra-high pressure liquid chromatography (UHPLC) is often used in tandem with a high-resolution mass spectrometer. Mobile phase composition, especially trifluoroacetic acid (TFA) concentration, is another important factor. On one hand, increasing TFA concentration to 0.1% maximizes chromatographic resolution, but on the other, comes at the cost of MS sensitivity. For maximal MS sensitivity, 0.1% formic acid is routinely used in the mobile phase in the absence of TFA. The drawback of formic acid is its strong UV absorbance, which interferes with and precludes the use of the UV chromatogram for peptide quantitation. A lower TFA concentration (~0.02%) is often used as a compromise. A UV chromatogram is useful in cases when an abundant sequence variant is not identified by MS/MS. It can also be used to provide a more accurate quantitation in cases where the peptides of interest are chromatographically resolved from other peptides.

7.3. MS and MS/MS

High-resolution mass analyzers such as an Orbitrap or a time-of-flight system are required to identify and quantify low-level sequence variants due to their superior resolving power and mass accuracy. In tandem MS, each peptide ion is isolated and further fragmented by collision-induced dissociation or another fragmentation technique to collect amino acid sequence information (Steen and Mann, 2004). In a typical LC-MS/MS experiment, a full MS scan is acquired in the high-resolution mode, followed by several data-dependent MS/MS scans at the same or lower resolution. Although low-resolution instruments have previously demonstrated successes (Yang et al., 2010), a high-

resolution mass spectrometer is generally required for confident sequence variant analysis. Full-scan MS are acquired at the highest possible resolution, while maintaining appropriate sensitivity and acquisition speed. The number of MS/MS scans is set appropriately to achieve at least six full MS scans across a chromatographic peak (Zhang et al., 2013). This ensures that a sufficient number of full-scan data points across the peak are available for integration and quantitation, and for distinguishing between closely eluting isomeric peptides.

To identify the maximum number of components in a complex mixture by data-dependent MS/MS, a dynamic precursor ion exclusion technique is needed in order to acquire a large number of unique MS/MS. This technique is important to achieve both breadth and depth for low-level sequence variants. The conceptual basis for dynamic exclusion is that once the MS/MS scan of a precursor ion is acquired, additional MS/MS of the same precursor ion is no longer required for identification. By temporarily assigning the m/z value of the ion to an exclusion list, the instrument can then efficiently allocate more time towards acquiring the MS/MS of other, less abundant ions. To ensure the acquisition of at least one high-quality MS/MS for each precursor ion, an exclusion duration of 2/3 to 3/4 of a typical chromatographic peak width is suggested (Zhang et al., 2013).

To achieve an even greater number of unique MS/MS, an inter-sample data-dependent acquisition strategy has been developed and used to analyze multiple similar samples (Zhang, 2012a). A natural evolution of dynamic exclusion, this strategy is based on the concept that once the MS/MS of a precursor ion is collected in one sample, its collection is no longer needed for identification in the subsequent similar samples. This approach further frees up instrument resources, so that a significantly greater number of sequence variants and other protein modifications can be identified (Zhang, 2012a, 2013). Lastly, while collision induced dissociation (CID) or higher-energy collisional dissociation (HCD) are typically used for collecting MS/MS data, when necessary, sequence variants can be confirmed by alternative fragmentation techniques such as electron-transfer dissociation (ETD).

7.4. Data analysis

LC-MS/MS analysis of a protein digest on modern high-resolution instruments provides a large amount of information regarding the structural attributes of the protein (Shah et al., 2014; Zhang, 2009; Zhang et al., 2009, 2013). Extracting all of the information contained in the LC-MS/MS data can be challenging. Recently, significant efforts have been made in creating informatics tools for this task. These tools are able to identify and quantify a large number of sequence variants and other structural attributes (Kil et al., 2015; Shah et al., 2014; Zhang, 2009; Zhang et al., 2013). Most of these tools match the determined mass and fragmentation pattern of a peptide to the sequence of theoretically digested peptides. Early sequence variant analysis was performed by error-tolerance search using Mascot (Borisov et al., 2015; Harris et al., 2012; Ragionieri et al., 2015; Ren et al., 2011; Yang et al., 2010; Yu et al., 2009; Zeck et al., 2012). MassAnalyzer (available from Thermo-Fisher Scientific as PepFinder™ or Biopharma Finder™) is able to assign identity and quantify detected sequence variants in an automated fashion (Brady et al., 2015; Traylor et al., 2016; Zhang et al., 2013). Similar software tools include Byologic/Byomap from Protein-Metrics (Kil et al., 2015) and Expressionist from Genedata (<https://www.genedata.com/products/expressionist/>). For illustration purposes, the sequence variant analysis workflow using MassAnalyzer is described below.

MassAnalyzer processes all data files for samples of the same protein

sequence simultaneously. As the first step, MassAnalyzer performs feature extraction on the data from each sample (Zhang, 2009) to detect all ions (retention time and isotope distribution) above a user-defined intensity threshold. Detected ions from each sample are then aligned to the reference sample data (the most representative sample dataset as determined by MassAnalyzer) based on retention time using a divide-and-conquer algorithm (Zhang, 2012b). A gap-filling procedure also searches for features, below the user-defined threshold, that were present in one sample, but missing from another in the original alignment (Zhang, 2012b). By aligning features, sequence variant levels can be directly compared between samples, and more importantly, once a feature is identified in one sample, the identification is applied to all other samples. Peptides are identified by first searching the determined monoisotopic mass against the protein sequence for potential candidates. The theoretical MS/MS of each candidate is then predicted with a kinetic model (Zhang, 2004, 2005, 2010, 2011; Zhang and Shah, 2010) and compared to the observed spectrum. A positive peptide identification is established when the predicted spectrum matches the observed spectrum. In a similar manner, potential sequence variants are proposed based on the determined mass and then the observed MS/MS is compared to the predicted MS/MS of proposed sequence variants for identification.

Sequence variant abundance is determined by comparing the peak areas under the selected ion chromatograms of the variant peptide and the native peptide. To minimize interference, a matched window function is used to generate the selected ion chromatogram (Zhang, 2009). Sequence variant quantitation is based on the assumption that the amino acid substitution does not alter the ionization efficiency of the peptide. This assumption is appropriate for many amino acid substitutions, but may not be valid for modifications involving large changes in basicity (changes involving introduction or elimination of a basic moiety) or hydrophobicity. Quantitation errors can also occur when a substitution either removes or creates a proteolytic cleavage site. This type of substitution changes the digestion pattern, and therefore peptides with different lengths, and likely different ionization efficiencies, are used for the quantitation. In general, quantitated levels are reproducible and linear, when several samples are processed and analyzed in parallel. This relative quantitation approach is useful when comparing relative amounts in different samples, and is fit-for-purpose when absolute quantitation is not critical.

7.5. Manual verification to remove false positives

The most significant challenge while analyzing LC-MS/MS data for sequence variants is the prevalence of potential false positives in the output of an informatics tool. False positive rates from informatics tools can sometimes be higher than 80% (Yang et al., 2010). These false positives are primarily due to misinterpreting chemical modifications as sequence variants, where the modifications have the same mass change as an amino acid substitution. For example, radical induced oxidation affects many amino acid side chains, leading to misidentification as Ala → Ser or Phe → Tyr sequence variants. Another example is the misidentification of carboxymethylation of methionine, histidine, lysine, N-terminal amine or carboxymethylated-cysteine by iodoacetic acid as Gly → Asp or Ala → Glu sequence variants (Zhang et al., 2013). Terminal amino acid additions caused by protease-catalyzed transpeptidation (Schaefer et al., 2005) are also frequently misidentified as sequence variants. Table 4 lists some frequently observed false positives, most of which have been reported previously by our group (Zhang et al., 2013). Careful manual examination of the MS/MS is necessary in order to distinguish between false positives and true sequence variants. Knowledge of protein chemistry, peptide fragmentation behavior, combined with consideration of retention times and other factors, are particularly useful to this end. In MassAnalyzer/Pep Finder/Biopharma Finder, peptides are identified by comparing the experimental MS/MS to accurately predicted theoretical MS/MS (Zhang, 2004, 2005, 2010,

Table 4

Frequently observed false positives and their causes in LC-MS/MS-based sequence variant analysis, assuming peptide mass changes can be determined to within ± 0.03 Da.

False positive (one-base change)	False positive Δ mass	Cause/modification	True modification site ^a	Modification Δ mass
A → D	43.990	Double Na adduct	Nonspecific	43.964
A → E	58.005	Carboxymethylation	C, H, M, K, Cm-C, N-term	58.005
A → G	− 14.016	CH ₂ loss	M, P, Cam-C	− 14.016
A → S	15.995	Oxidation	Many	15.995
D → A	− 43.990	Gas-phase CO ₂ loss	C-term, E, D	− 43.990
D → G	− 58.005	Incomplete carboxymethylation	Cm-C	− 58.005
D → N	− 0.984	Amide formation	C-term, Cm-C	− 0.984
E → A	− 58.005	Incomplete carboxymethylation	Cm-C	− 58.005
E → D	− 14.016	CH ₂ loss	M, P, Cam-C	− 14.016
E → Q	− 0.984	Amide formation	C-term, Cm-C	− 0.984
F → V	− 48.000	CH ₃ S loss from Met	M	− 48.003
F → Y	15.995	Oxidation	Many	15.995
G → D	58.005	Carboxymethylation	C, H, M, K, Cm-C, N-term	58.005
H → N	− 23.016	Oxidation of His to Asn	H	− 23.016
I → N	0.959	Deamidation	N, Q, Cam-C	0.984
I → R	43.017	Carbamylation	N-term, K	43.006
I → V	− 14.016	CH ₂ loss	M, P, Cam-C	− 14.016
K → R	28.006	Formylation	N-term, S, T	27.995
L → R	43.017	Carbamylation	N-term, K	43.006
L → V	− 14.016	CH ₂ loss	M, P, Cam-C	− 14.016
M → T	− 29.993	Gas-phase H ₂ CO loss	C-term, S, T	− 30.011
N → D	0.984	Deamidation	N, Q, Cam-C	0.984
N → I	− 0.959	Amide formation	C-term, Cm-C	− 0.984
P → A	− 26.016	Double oxidation of Cys	Cm-C	− 26.016
P → S	− 10.021	Triple oxidation of Cys	Cm-C	− 10.021
P → T	3.995	Oxidation of Trp to kynurenine	W	3.995
Q → E	0.984	Deamidation	N, Q, Cam-C	0.984
R → K	− 28.006	C ₂ H ₄ loss	P	− 28.031
R → Q	− 28.043	C ₂ H ₄ loss	P	− 28.031
R → W	29.978	Oxidation product of Tyr	Y	29.974
S → G	− 30.011	Gas-phase H ₂ CO loss	C-term, S, T	− 30.011
T → A	− 30.011	Gas-phase H ₂ CO loss	C-term, S, T	− 30.011
T → M	29.993	Oxidation product of Tyr	Y	29.974
T → S	− 14.016	CH ₂ loss	M, P, Cam-C	− 14.016
V → A	− 28.031	C ₂ H ₄ loss	P	− 28.031
V → E	29.974	Oxidation product of Tyr	Y	29.974
V → F	48.000	Triple oxidation	W, C	47.985
V → M	31.972	Double oxidation	M, W, C, Y	31.990
Y → H	− 26.004	Double oxidation of Cys	Cm-C	− 26.016
X → D, E, K, R, etc.	Many	Transpeptidation ^b	N-term, C-term	Many

^a Cm-C: carboxymethylated cysteine; Cam-C: carbamidomethylated cysteine.

^b Asp, Lys, Glu and Arg residues are added to the N- or C-terminus through protease-catalyzed transpeptidation by Asp-N (Asp), Lys-C (Lys), Glu-C (Glu) or trypsin (Lys and Arg), etc.

2011; Zhang and Shah, 2010), thereby providing a more reliable identification platform without the need for users to understand peptide fragmentation behavior. MassAnalyzer also removes false positives based on Table 4 and other important factors such as retention time. A typical MassAnalyzer output (amino acid substitutions with a single-base difference in their codons; sequence variant detection threshold set to 0.001%) contains approximately 50% confirmed true positives, approximately 10% confirmed false positives, and the remaining

unconfirmed results being caused by insufficient information in the MS/MS.

The following considerations can provide supporting evidence when manually confirming true sequence variants:

- 1) Some sequence variants involve a residue that either creates or removes a proteolytic cleavage site. These variants can often be confirmed by a digestion pattern change with the corresponding protease. Examples include Gly → Asp for Asp-N, Gly → Glu for Glu-C, Asn → Lys for trypsin and Lys-C, Ser → Arg and Arg → Gln for trypsin, etc. Protease-catalyzed transpeptidation should be ruled out first.
- 2) Different proteases can create alternative digestion patterns. The MS/MS of the alternative digest may provide additional information and cross-confirmation towards sequence variant identity.
- 3) Other fragmentation techniques such as electron-transfer dissociation (ETD) can provide a different fragmentation pattern, which can facilitate deduction of the sequence variant identity and location.
- 4) Some amino acid substitutions cause a significant peptide hydrophobicity change, which will be reflected in their retention times on reversed-phase columns.
- 5) When necessary, a sequence variant can be confirmed by Edman sequencing. Alternatively, a synthesized peptide can be used as a reference to confirm the sequence variant by comparing against retention time and MS/MS pattern. DNA mutations are confirmed by DNA sequencing.

7.6. DNA mutation vs. amino acid misincorporation

Sequence variants caused by DNA mutations can pose risks to biopharmaceutical products. They should be distinguished from amino acid misincorporation products and other modifications during clone selection, to prevent their manifestation in the protein product. This can be accomplished by comparing data between different clones. A mutation-derived sequence variant is clone specific, and will only appear at the mutation site, whereas amino acid misincorporation products are found broadly across the entire protein sequence (Zhang et al., 2013). A workflow, based on similar principles, can also be applied to distinguish between mutation-derived sequence variants and false positives (Brady et al., 2015; Zeck et al., 2012). Although DNA mutation screening is ideally performed at the genetic and transcript level, protein level screening by LC-MS/MS offers the advantage of high sensitivity, as well as additional information on translational and post-translational variants in a single analysis. DNA mutations detected by LC-MS/MS are often confirmed by DNA sequencing. In a biopharmaceuticals context, however, DNA mutation confirmation is rarely necessitated as the corresponding cell lines often fail to reach the later stages of drug development.

7.7. Strengths and limitations of the LC-MS/MS technique

The major advantage of using LC-MS/MS for sequence variant analysis is its capability to broadly identify amino acid substitutions in a single analysis. The limited exceptions are substitutions that are indistinguishable by mass alone from chemical modifications such as, deamidation of Asn (Asn → Asp), deamidation of Gln (Gln → Glu), etc., or product isomers (Leu and Ile substitution products, Tyr isomers, etc.). Additionally, mass spectrometry methods afford high sensitivity. With current technology, amino acid substitutions with levels approaching 0.001% may be identified (Zhang et al., 2013).

Limitations of the LC-MS/MS technique for sequence variant analysis are listed below.

- 1) As false positives can be prevalent, conclusions from LC-MS/MS should be approached with thorough consideration. For definitive confirmation, techniques such as Edman sequencing or LC-MS/MS

confirmation with a synthetic reference peptide are used. DNA mutations can be confirmed by DNA or RNA sequencing.

- 2) Since an amenable peptide size for LC-MS/MS is between 500 and 3000 Da, effective sequence coverage from peptide mapping can be incomplete. For example, for a tryptic digestion of an antibody, only approximately 70% of the entire sequence is within the desired range. Amino acid substitutions occurring in the other 30% of the molecule are likely undetectable by trypsin digestion alone.
- 3) Even with sufficient MS/MS coverage, LC-MS/MS is limited to the identification of products that are distinguishable by mass.
- 4) The chance of identifying a sequence variant is determined by many factors such as the length of the proteolytic peptide, peptide separation resolution, and MS/MS quality, to name a few. As a result, although sequence variants at 0.001% are routinely detected, the practical detection limit is usually higher. For example, it has been estimated that when a single protease (trypsin) was used to digest a mAb, only approximately 30% of sequence variants within the 0.01–0.1% range were identified (Zhang et al., 2013).
- 5) Sequence variant quantitation is based on the assumption that the variant peptide and native peptide have a similar MS ionization response. The assumption is valid in many cases. However, substitutions that significantly alter digestion patterns, basicity, and peptide hydrophobicity are likely to be quantitated with limited accuracy.

8. Conclusions

Amino acid misincorporation during protein expression, resulting in the formation of unintended protein sequence variants, has been known to occur since the 1960's. However, its causes, mechanisms, exacerbating factors, and impact are still being heavily studied today. The study of this phenomenon has become increasingly important to the biopharmaceutical industry where high product quality is required to ensure patient safety and product efficacy. Root causes of unintended protein sequence variants are found in all three information propagation points in the protein biosynthesis machinery, including transcription, tRNA charging and codon-anticodon base pairing. The formation of sequence variants has been linked to codon usage, cell metabolism, cellular stress, and nutrient starvation. Due to the complex nature of the phenomenon, efforts to minimize amino acid misincorporation will likely require a concerted effort at the genetic, translational, metabolic, cellular, and process levels. Current methods used to identify and characterize cases of amino acid misincorporation take advantage of the high-resolution tandem-mass spectrometry methods. These quickly evolving methods provide researchers the ability to seamlessly determine protein primary sequence, elucidate structural attributes, identify instances of amino acid misincorporation, and characterize post-translational modifications concurrently. Continued advances in mass spectrometry technology, with respect to speed, sensitivity, and resolution, along with the implementation of existing and emerging automated data analysis tools will continue to improve the accuracy and efficiency of LC-MS/MS techniques.

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